INCH-POUND
MIL-P-70877 (AR)
22 August 1991

MILITARY SPECIFICATION

PRINTED WIRING BOARD ASSEMBLY, ROCKET TRIGGER INTERFACE, SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the Printed Wiring Board Assembly (PWBA), Rocket Trigger Interface, part number 9358750.

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	- Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1686	- Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	- Environments, Working, Minimum Standards for
MIL-STD-2000	- Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	- Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

9358750	-	Printed Wiring Board Assy, Rocket Trigger Interface
9358752	-	Schematic Diagram, Rocket Trigger Interface

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P9358750	-	Printed Wiring Board Assy, Rocket Trigger Interface

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The Rocket Trigger Interface Printed Wiring Board Assembly performs the function of rocket inventory control and rocket trigger generation. Inventory may initially be set to any value between zero and nineteen. Each firing decreases the available stores by one and outputs a rocket trigger. A separate line for each rocket is grounded for about two seconds at system turn on if that particular rocket has been put in stores. As a rocket is fired, the line will no longer be grounded. When all inventory is fired, the rocket trigger signal line will no longer output a signal.

These functions are implemented using shift registers to keep track of stores and relays to ground the individual rocket lines. Miscellaneous random logic and discrete semiconductors are also employed on this printed wiring board.

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°c and +63°c. After exposure to ESS in accordance with 4.5, the characteristics of 3.4 shall not be degraded.

3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table I, measured under quiescent conditions. For terminal and component designations called out in this specification, see drawing 9358752, Schematic Diagram, Rocket Trigger Interface.

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TABLE I. Operating Conditions.

Parameter	Terminal	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-32	25	63	°C
Input (+)	J4-8	7.3	11.0	12.5	Vdc
Power (+)	J4-11	11.75	12.0	12.25	Vdc
Return (-)	J4-1				
Power (-)	J3-1				
Input	J4-8	0.10	0.2	235	mA 1/
Current	J4-11	4.0	8.0	40.0	mA 1/

NOTE:
1/ The maximum current shown is with all MOSFET transistors turned on. Quiescent current maximum is 5.0 mA for J4-8 and 10mA for J4-11.

3.4.2 Performance. Unless otherwise specified, the PWBA shall meet the requirements of this specification under the following conditions:

a. Temperature 25°±10°C (77°±18°F)
b. Relative humidity 50±30 %
c. Barometric pressure 725+50, -75mm Hg (28.5+2.0, -3.0 inches Hg)
d. Input power and loads See Figure 1

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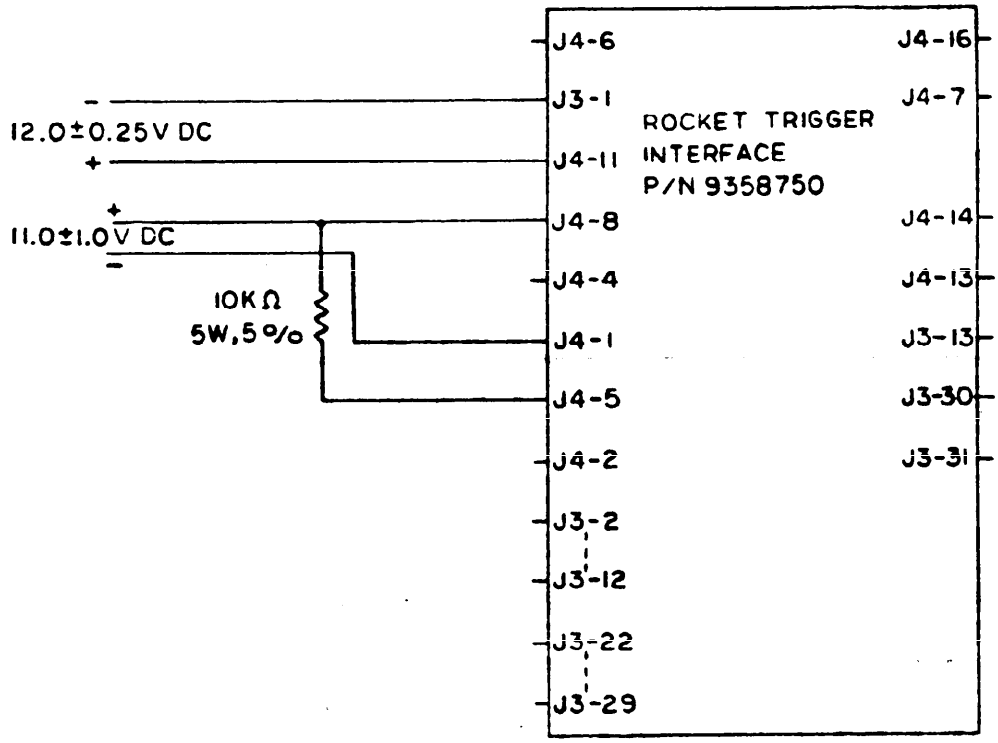


FIGURE 1. Rocket Trigger Interface PWBA Power Source and Loads.

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3.4.2.1 Current drain. With power applied to J4-8 and J4-11, the maximum quiescent current drain from the PWBA shall be 5mA and 12mA, respectively.

3.4.2.2 Power-on. When power is applied, an 11.0 ± 0.1V pulse shall be present across J4-6 (+) and J4-16 (-) for 3.0 ± 1.0 sec, then drop to zero; 200±35 mA shall flow into terminal J4-8 for the first 3.0 ± 1.0 sec then drop to less than 5.0 mA and 30 ± 10 mA shall flow into terminal J4-11 for the first 3.0 ± 1.0 sec then drop to 8 ± 4 mA.

Timer U12 provides an 11V pulse, once power is applied, to terminal J4-6 for 3 sec and lightly turns on 19 MOSFET transistors (with a gate-source voltage of 2V as opposed to 10V for full on) in U2, U3, U14, U15 and U24 (not including transistor 4 from U24), due to the diode reverse leakage current flowing through the 220 kohm resistors. With all 19 transistors on, current is sent through the 4th transistor of U24 and across the 3.3 ohm R20 resistor, since transistor Q1 is turned on with any of the 19 MOSFETS activated. With Q1 on and 19 MOSFETS on, 0.7V is put across R20 which pulls 210 mA from terminal J4-8. When the 11V pulse is provided by the timer, 10V is put across R10, which pulls 22 mA from terminal J4-11.

3.4.2.3 Fuse. With power applied, the resistance across J3-30 and J3-31 (the fuse circuit) shall be less than 2.0 ohms.

3.4.2.4 Inventory. With power applied, and J4-6 (inventory override) connected to J4-8 (but not more than 1 minute in any 3 minute period), the resistance at rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29 relative to J3-13 shall be 100 ohms minimum.

With the inventory override applied, the 19 MOSFET transistors are lightly turned on (with a gate-source voltage of 2V as opposed to 10V for full on) due to the diode reverse leakage current flowing through the 220 kohm resistors. The drain-source resistance of each MOSFET is then greater than 100 ohms which makes the resistance from the rocket sense/fire terminals to J3-13 greater than 100 ohms.

3.4.2.5 Inventory override. With power applied, J4-6 (inventory override) connected to J4-8 (but not more than 1 minute in any 3 minute period), and J4-2 (controller key) connected momentarily to J4-8, the resistance at rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29 relative to J3-13 shall be 10.0 ohms maximum.

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With the controller key applied, set/reset latches U5, U6, U17, U18 and U25 are all set high, which fully turns on the 19 MOSFETS (with a gate-source voltage of 10V) since inventory override is present. The drain-source resistance of the 19 MOSFET transistors is then approximatley 0.7 ohms, which, along with resistor R20, makes the resistance from the rocket sense/fire terminals to J3-13 less than 10 ohms.

3.4.2.6 Rocket trigger. With power applied, and J4-8 sequentially connected to rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29, a rocket trigger pulse of 11.5 ± 0.5 V dc shall appear across J4-7(+) (aircraft trigger LED) and J4-14(-) and a positive pulse of 11.0 ± 0.5 V dc shall appear across J4-13(+) (rocket trigger) and J3-13(-) while each connection is made to J4-8.

When one of the rocket sense/fire lines is pulled high, one of the MOSFET transistors from U2, U3, U14, U15 or U24 is turned on and produces a high pulse at J4-7, since the drain side from all the MOSFET transistors are tied to J4-7 through a diode. The rocket sense/fire line also sets one of the set/reset latches from U5, U6, U17, U18 or U25 which disables "nand" gate U11, U22 or U26 (pin 6). This then enables U26 (pin 9), which disables U26 (pin 10) and produces a high pulse at J4-13 while the rocket sense/fire is high.

3.4.2.7 Fired inventory. With power applied, J4-8 sequentially connected to rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29 and J4-6 (inventory override) connected to J4-8 (but not more than 1 minute in any 3 minute period), the resistance at J3-2 through J3-12 and J3-22 through J3-29 relative to J3-13 shall be 100 ohms minimum.

With the inventory override applied, the 19 MOSFET transistors are lightly turned on (with a gate-source voltage of 2V as opposed to 10V for full on) due to the diode reverse leakage current flowing through the 220 kohm resistors. When the 19 rocket sense/fire terminals are pulled high, the set/reset latches U5, U6, U17, U18 and U25 are reset, but do not affect the MOSFETs gate-source voltages. The drain-source resistance of each MOSFET is then greater than 100 ohms which makes the resistance from the rocket sense/fire terminals to J3-13 greater than 100 ohms.

3.4.2.8 Reset inventory. With power applied, J4-8 sequentially connected to rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29, J4-6 (inventory override)

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connected to J4-8 (but not more than 1 minute in any 3 minute period) and J4-2 (controller key) momentarily connected to J4-8, the resistance at J3-2 through J3-12 and J3-22 through J3-29 relative to J3-13 shall be 10.0 ohms maximum.

With the controller key applied, set/reset latches U5, U6, U17, U18 and U25 are set high, which, along with the inventory override, fully turns on the 19 MOSFETS (with a gate-source voltage of 10V). The rocket sense/fire terminals do not effect the set/reset latches. The drain-source resistance of the 19 MOSFET transistors is then approximatley 0.7 ohms, which makes the resistance from the rocket sense/fire terminals to J3-13 less than 10 ohms.

3.4.2.9 Rocket test. With power applied, J4-8 sequentially connected to rocket sense/fire terminals J3-2 through J3-12 and J3-22 through J3-29, J4-6 (inventory override) connected to J4-8 (but not more than 1 minute in any 3 minute period), J4-2 (controller key) momentarily connected to J4-8 and J4-5 (rocket test) momentarily connected to J4-1 19 times, the resistance at J3-2 through J3-12 and J3-22 through J3-29 relative to J3-13 shall change from less than 10.0 ohms to greater than 100 ohms in sequence with each J4-5 connection and an 11.5 $\pm$ 0.5 Vdc pulse shall be output at J4-13(+) and J3-13(-) each time J4-5 and J4-1 are connected.

When the rocket sense/fire terminals are pulled high each set/reset latch is reset. The inventory override line then lightly turns on all 19 MOSFETs (the gate-source drain voltage will be approximately 2V for each MOSFET instead of 10V, required for full turn on). After the controller key is applied, the set/reset latches are set which will put the 19 MOSFET transistors full on (with a gate-source voltage of 10V). With the 19 MOSFETs fully on, the 4th MOSFET at U24 is full on, which makes resistor R20 the largest resistance from each rocket sense/fire line to J3-13. The rocket sense/fire resistance is then less than 10 ohms. When the rocket test terminal is pulled low 19 times, the dual 4 stage static shift registers U8, U20 and U23 are sent clock signals which sequentially sends a high to each rocket sense/fire line and resets the corresponding set/reset latch. When a set/reset latch is reset, "nand" gate U11, U22 or U26 (pin 6) is disabled, which enables U26 (pin 9), which disables U26 (pin 10), which puts a high pulse at J4-13. Also, when the set/reset latches are sequentially reset, the corresponding MOSFET transistor is changed from fully on (gate-source voltage of 10V) to lightly on (gate-source voltage 2V). This will raise the drain-source resistance of the MOSFET which

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will raise the corresponding rocket sense/fire resistance from less than 10 ohms to greater than 100 ohms.

3.4.2.10 Tow fire. With power applied, and J4-4 (tow fire) momentarily connected to J4-8, a positive pulse of 11.0±0.5 Vdc shall be present at J4-7(+) (aircraft trigger) and J4-14(-) while J4-4 and J4-8 are connected. A voltage of 0.5±0.5 Vdc shall be present at J4-7(+) and J4-14(-) after J4-4 and J4-8 are disconnected.

The TOW fire and actual trigger terminals are connected internally by diode CR35 and resistor R11. Therefore, a pulse at the TOW fire terminal results in a pulse at the aircraft trigger terminal.

3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subject to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2. when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +55°C (+131°F).

3.4.3.1.2 Storage temperature. The assembly shall operate after exposure to a temperature of +63°C (+145°F) for a period of 48 hours.

3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -35°C (-30°F)for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 2. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

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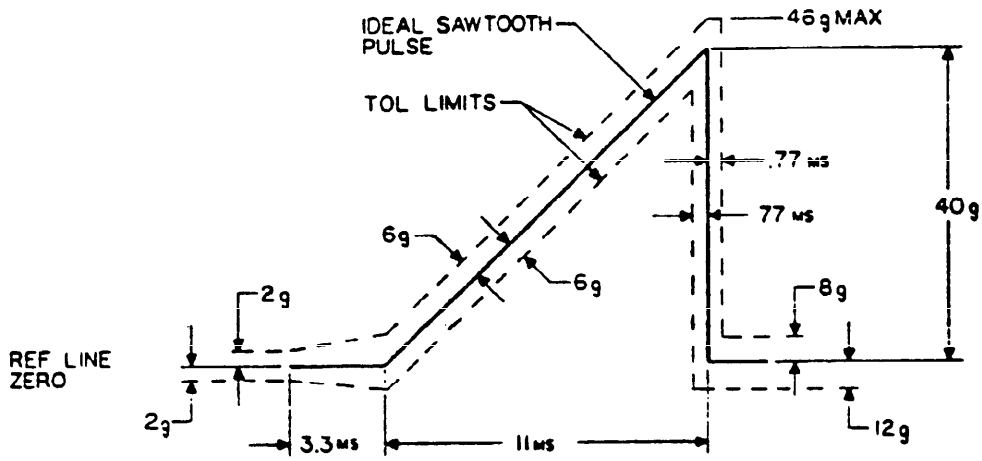


FIGURE 2. Shock Pulse.

3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 3 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

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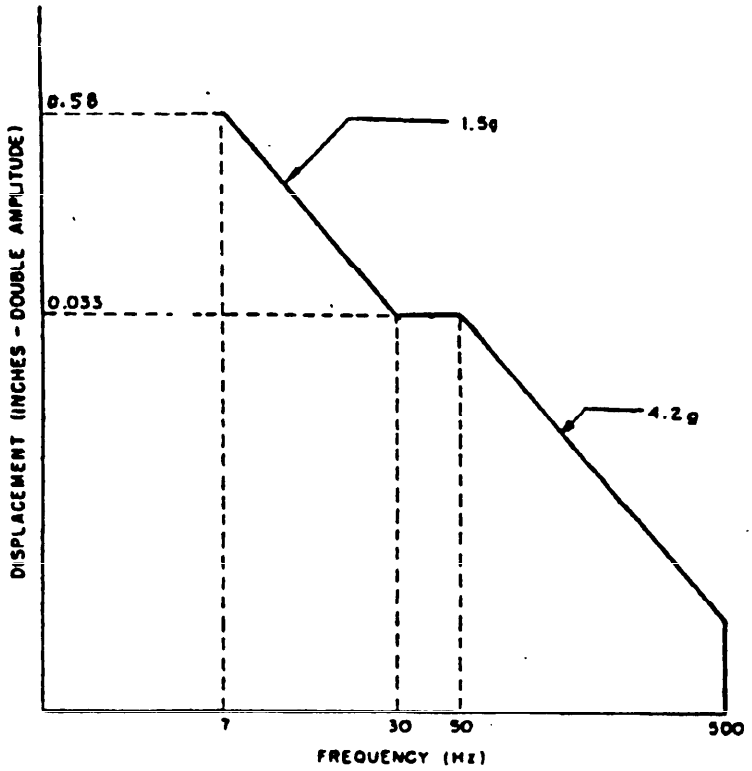


FIGURE 3. Vibration Test.

3.5 Design and construction. The PWBA shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawing 9358750. The PWBA shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

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3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental stress screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in Tables II and III. Each first article sample unit shall be inspected in accordance with the Table II before being subjected to the environmental tests in Table III in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE II. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.1.1	3	32/Acc-0,Rej-1
102	Power-on	3.4.2.2	4.8.1.2	3	32/Acc-0,Rej-1
103	Fuse	3.4.2.3	4.8.1.3	3	32/Acc-0,Rej-1
104	Inventory	3.4.2.4	4.8.1.4	3	32/Acc-0,Rej-1
105	Inventory Override	3.4 .5	4.8.1.5	3	32/Acc-0,Rej-1
106	Rocket Trigger	3.4 .6	4.8.1.6	3	32/Acc-0,Rej-1
107	Fired Inventory	3.4.2.7	4.8.1.7	3	32/Acc-0,Rej-1
108	Reset Inventory	3.4.2.8	4.8.1.8	3	32/Acc-0,Rej-1
109	Rocket Test	3.4.2.9	4.8.1.9	3	32/Acc-0,Rej-1
110	TOW Fire	3.4.2.10	4.8.1.10	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coating	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 (major characteristics) class performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE III. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size/ Accept Criteria
111	High Temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
112	Low Temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
113	Shock	3.4.3.3	4.9.3	3	1/
114	Vibration	3.4.3.4	4.9.4	3	1/

NOTE:
1/ Not required for quality conformance inspection.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be sujected to a thermal stress screen in accordance with Table IV and the procedure below.

TABLE IV. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/Group A Test
-35°C	+63°C	15°C/minute	Off	No	20/ Failure Free or Additional Cycle

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4.5.1 Conformal coating. Thermal stress screen shall be performed prior to conformal coating unless another method is approved by the Government.

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -35°C + 63°C (-30°F to 145°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 4).

4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted with units nonoperating and without power (static).

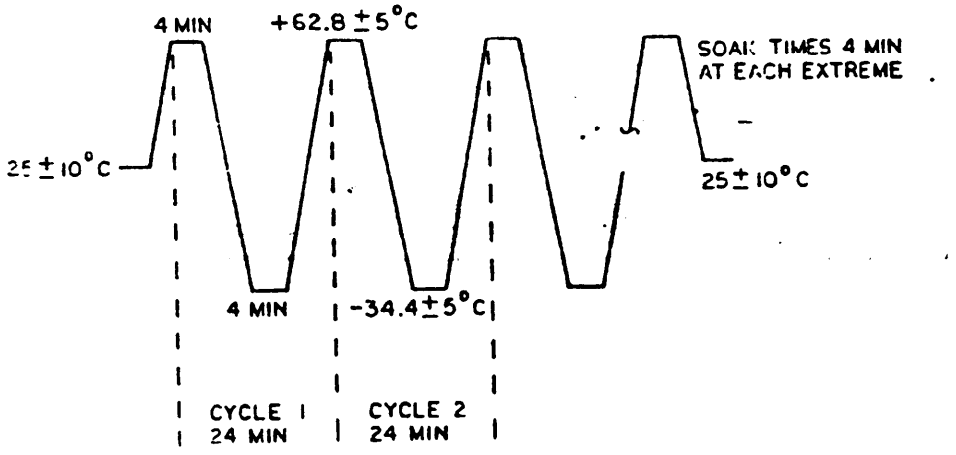


FIGURE 4. Environmental Stress Screen Temperature Profile.

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.5.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from

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which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table II in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table III and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements of 5.1. During the period of first article inspection, one additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

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4.6.5 Failure of sample. Should any one sample fail to meet any of the specified examinations and test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.7 and 4.8 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test

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methods and procedures as specified in 4.8.

TABLE V. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1.
DC Voltmeter/Multimeter	Standard - capable of measuring voltages stated in 4.8.1.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.1.
Pulse Generator	Standard - capable of providing pulses stated in 4.8.1.
Oscilloscope	Standard - capable of measuring signals stated 4.8.1.
Ohmmeter	Standard - capable of measuring resistances stated in 4.8.1.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Performance tests. The following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The PWBA shall be connected to the circuit specified in Figure 5. All switches shall remain as shown in Figure 5 prior to each test specified in 4.8.1.1 through 4.8.1.10.

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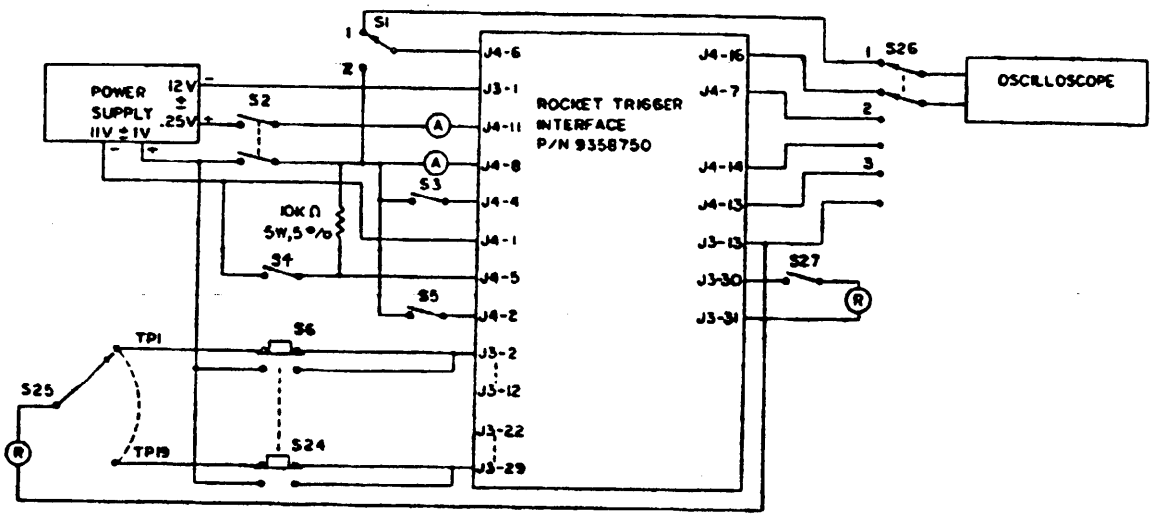


FIGURE 5. Test Circuit for Rocket Trigger Interface.

4.8.1.1 Current drain. Apply power by turning on S2 and verify the current drain requirements of 3.4.2.1.

4.8.1.2 Power-on. Set S26 to position 1, turn ON S2 and

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verify the power-on requirements of 3.4.2.2.

4.8.1.3 Fuse. Turn ON S27 and verify the fuse requirement of 3.4.2.3 with ohmmeter at S27.

4.8.1.4 Inventory. Turn ON S2, set S1 to position 2 (do not leave S1 in position 2 for more than 1 minute in any 3 minute period), set S25 to TP1 through TP19 and for each S25 position verify the inventory requirements of 3.4.2.4 with the ohmmeter.

4.8.1.5 Inventory override. Turn ON S2, set S1 to position 2 (do not leave S1 in position 2 for more than 1 minute in any 3 minute period), turn ON S5 momentarily, set S25 to TP1 through TP19 and for each S25 position, verify the inventory override requirements of 3.4.2.5 with the ohmmeter.

4.8.1.6 Rocket trigger. Turn ON S2, set S26 to position 1 and sequentially turn ON S6 through S24 while verifying the rocket trigger requirements of 3.4.2.6 with the oscilloscope. Repeat the above procedure with S26 set to position 3.

4.8.1.7 Fired inventory. Turn ON S2, set S1 to position 2 (do not leave S1 in position 2 for more than 1 minute in any 3 minute period), turn ON S6 through S24, each momentarily, and verify the fired inventory requirements of 3.4.2.7 for S25 at positions TP1 through TP19 with the ohmmeter.

4.8.1.8 Reset inventory. Turn ON S2, set S1 to position 2 (do not leave S1 in position 2 for more than 1 minute in any 3 minute period), turn ON S6 through S24, each momentarily, turn ON S5 momentarily and verify the reset inventory requirements of 3.4.2.8 for S25 at positions TP1 through TP19 with the ohmmeter.

4.8.1.9 Rocket test. Turn ON S2, set S1 to position 2 (do not leave S1 in position 2 for more than 1 minute in any 3 minute period), turn ON S6 through S24, each momentarily, turn ON S5 momentarily, set S26 to position 3 and turn ON S4 momentarily 19 times. Verify the rocket test requirements of 3.4.2.9 at S25 in positions TP1 through TP19 with the ohmmeter and at S26 with the oscilloscope each time S4 is turned ON.

4.8.1.10 TOW fire. Turn ON S2, set S26 to position 2, momentarily turn ON S3 and verify the requirements at 3.4.2.7 with the oscilloscope.

4.9 Environmental tests.

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a. Unless otherwise specified, environmental tests shall be conducted in accordance with test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the test required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501 using the following procedure:

a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

b. Step 2 - Raise the internal temperature to 63°C (145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 48 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Adjust the internal chamber temperature to 55°c (131°F), the highest operating temperature for which the test item is designed to operate, and maintain until temperature stabilization of the item is reached.

e. Step 5 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

f. Step 6 - Return the test item, nonoperating, to standard ambient condition and stabilize.

g. Step 7 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.2 Low temperature. This satisfies the storage and

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operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain the temperature for period of 4 hours.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.3 Shock. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 6), shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

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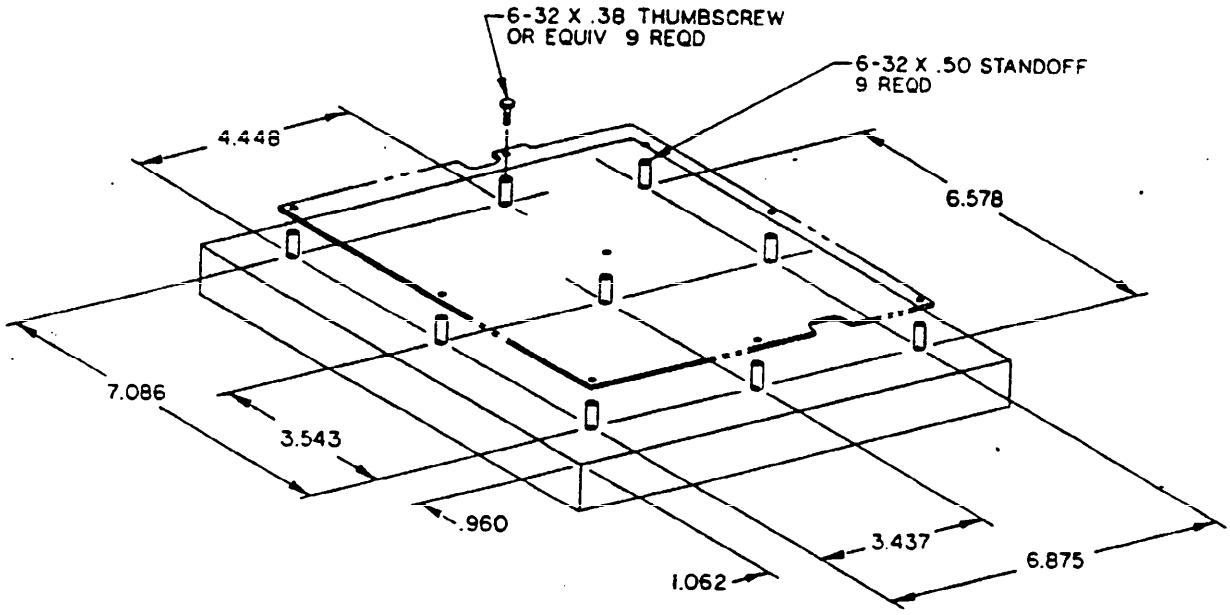


FIGURE 6. Mounting Configuration.

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4.9.4 Vibration. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 6), shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes per axis with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI-P9358750. The level of protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Rocket Trigger Interface PWBA is part of the Multiple Integrated Laser Engagement System.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF- I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 MILES. Multiple Integrated Laser Engagement System.

6.4.2 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.7 Subject term (key word) listing.

Fire Control
Training Devices

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A035)

User activities:
Army - MR
Navy - CG, MC
Air Force - 90

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70877 (AR)	22 August 1991

3. DOCUMENT TITLE: PWB ASSEMBLY, ROCKET TRIGGER INTERFACE, SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commander US Army ARDEC	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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